

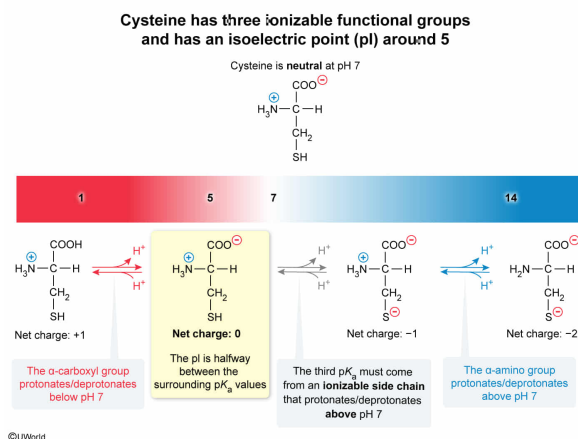
An unknown amino acid **X** was found to have three pK_a values, two above 7 and one below 7. Furthermore, its isoelectric point (pI) was determined to be below 7.0, and its side chain was determined to be predominantly uncharged at pH 7.0. Which of the following amino acids is most likely amino acid **X**?

- ☐ A. Glutamate [23%]
- ☐ B. Valine [11%]
- ☐ C. Lysine [28%]
- ✓ ☒ D. Cysteine [36%]

Correct

37% answered correctly

Explanation:



The identities of amino acids can be determined by their pK_a values and their isoelectric point (pI). The pK_a value of a functional group is equal to the pH value where its predominant protonation state switches between protonated ($pH < pK_a$) and deprotonated ($pH > pK_a$). In contrast, the isoelectric point (pI) is the pH at which the average charge of the population of amino acids is electrically neutral.

The presence of **three pK_a values** indicates that the amino acid has **three ionizable functional groups**. For **amino acids**, two of these groups must be the **α-amino** ($pK_a > 7$) and **α-carboxyl** ($pK_a < 7$) groups. The third must come from an **ionizable side chain**; therefore, amino acid **X** *must* have an ionizable side chain, and this side chain must have a pK_a that is greater than 7.

At pH 7, the α-amino and α-carboxyl groups are positively and negatively charged, respectively, and thus **neutralize each other**. Because the **side-chain pK_a must be above 7**, the side chain is predominantly **protonated at pH 7**. The question indicates that this protonated form of the side chain is **uncharged**. Furthermore, because the question states that the **pI is below 7.0**, the side chain must stay protonated and

electrically neutral at pH levels below 7, but it can become predominantly deprotonated and negatively charged at higher pH levels.

Of the given options, only the **thiol group** of cysteine is neutral when protonated at pH 7 and below but can be deprotonated at pH values above 7 to gain a negative charge; therefore, the most likely identity of amino acid **X** is cysteine.

(Choice A) Glutamate is an acidic amino acid. Its side chain contains a carboxylate that is **negatively charged** at pH 7 and protonated around pH 4 ($pK_a < 7$).

(Choice B) Valine does *not* have an ionizable side chain and therefore has only **two pK_a values**.

(Choice C) Lysine does have a side chain pK_a above 7.0, but it is *positively* charged at pH 7 and remains positively charged at lower pH values, resulting in $pI > 7$.

Educational objective:

The amino acid cysteine contains three ionizable functional groups with distinct pK_a values. The thiol side chain is neutral at pH 7 or below but can become deprotonated and negatively charged at higher pH values.

Time spent: 0 sec

Subject:
Biochemistry

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System:
Amino Acids and Proteins